**15** Express $0.4\dot{1}\dot{8}$ as a fraction.
You must show all your working.

Let x = 0.4181818...

10x = 4.181818...

1000x = 418.181818...

990x = 414, so x = 414/990 = 23/55.

**3 marks total**
Write recurring multiples
Subtract: 990x = 414
Simplify to 23/55

.....

**(Total for Question 15 is 3 marks)**

**16** (a) Rationalise the denominator of $\frac{22}{\sqrt{11}}$

Give your answer in its simplest form.

.....

**(2)**

(b) Show that $\frac{\sqrt{3}}{2\sqrt{3}-1}$ can be written in the form $\frac{a+\sqrt{3}}{b}$ where $a$ and $b$ are integers.

**(3)**

**(Total for Question 16 is 5 marks)**

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